

Automatic Camera Based BOOM BARRIER Better Than The IR Based

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This is a common sight at big buildings, such as shopping malls and 5-star hotels, where the entrance for staff and regular vendors has a boom barrier. The staff cars and other regular vehicles have to slow down and wait near an infra-red (IR) scanner, which reads the faint radiation of an IR tag pasted on the vehicle's windshield. The boom barrier opens if the tag is found in its database. The process is quite efficient and works unhindered. However, the only flip side is that the process does not work when there is no tag, or the tag is damaged, or it cannot be identified for some reason.

The system proposed here takes care of such problems. When a vehicle approaches the boom barrier, a speaker automatically asks the driver to slow down and stop briefly within four metres of the scanner.

A high-fidelity camera then reads the vehicle's registration number on its registration plate and checks whether it is there in its database. If the number is found in the database, the boom barrier opens automatically and the driver is told on the loudspeaker to proceed.

Whenever a new unregistered vehicle arrives, its registration number is updated in the database (if the visitor is expected to visit regularly henceforth) and the process keeps running. This system requires no IR tag, no pasting of the tag on vehicles' windshields, and no big disk antenna for an IR scanner at the gate. The process can

be further refined, so that the vehicles need not stop at all, by using a better camera and computer.

To achieve this, a powerful Raspberry Pi 4 computer is needed for this project. For measuring distance, the same image analytic can be deployed. But to make process simple, deployment of an HC-SR04 ultrasonic sensor is recommended.

BILL OF MATERIAL

Components	Quantity
Raspberry Pi 3 or Pi 4 computer with 4 / 8GB RAM	1
Raspberry Pi camera	1
HC-SR04 ultrasonic sensor	1
7-inch touchscreen for Raspberry Pi	1
5V, 2A DC power supply	1
Speaker	1



Fig. 1: Author's prototype

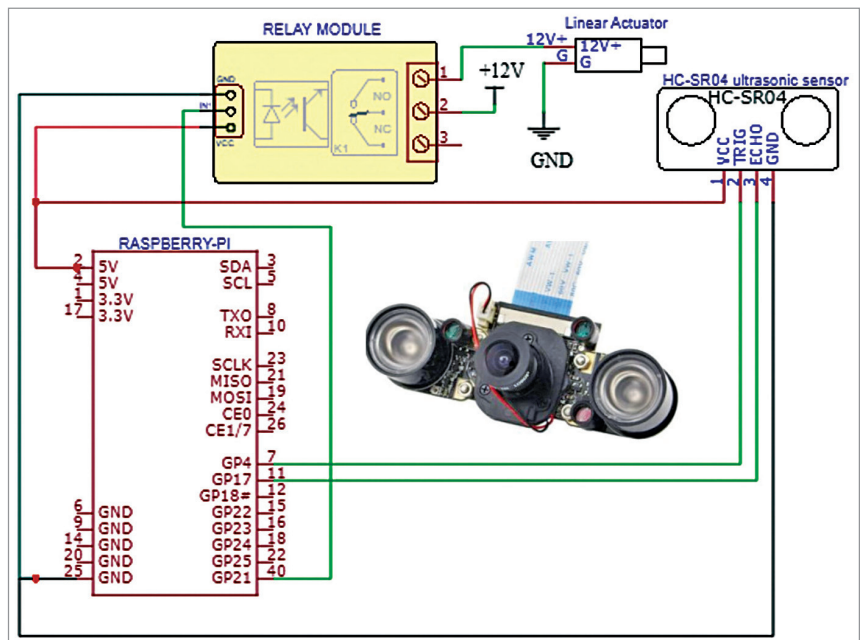


Fig. 2: Circuit diagram of project